

QMS Journal Club

Choong Pak Shen

February 22, 2026

QMS Journal Club 6th meeting: Schmidt
decomposition, bipartite entanglement, and state
purification

Higher order tensors

From one-qubit state

$$|\psi_A\rangle = \psi_0|0\rangle + \psi_1|1\rangle \quad (1)$$

to two-qubit state

$$|\psi_{AB}\rangle = \psi_{00}|00\rangle + \psi_{01}|01\rangle + \psi_{10}|10\rangle + \psi_{11}|11\rangle, \quad (2)$$

the probability amplitude of the quantum states changes from an one-index object to a two-index object.

In general, one-index object $\Psi_A = (\psi_i)$ is called tensor of order 1, while two-index object $\Psi_{AB} = (\psi_{ij})$ is called tensor of order 2.

Tensor of order 1 is usually represented as a column vector and it coincides with the state vector representation.

Meanwhile, tensor of order 2 is represented as a matrix. In two-qubit case,

$$\Psi_{AB} = (\psi_{ij}) = \begin{pmatrix} \psi_{00} & \psi_{01} \\ \psi_{10} & \psi_{11} \end{pmatrix}. \quad (3)$$

Schmidt decomposition

Theorem: Suppose $|\psi\rangle$ is a pure state of a composite system AB . Then there exist orthonormal states $|i_A\rangle$ for system A , and orthonormal states $|i_B\rangle$ for system B , such that

$$|\psi\rangle = \sum_i \lambda_i |i_A\rangle |i_B\rangle, \quad (4)$$

where λ_i are non-negative real numbers satisfying $\sum_i \lambda_i^2 = 1$, known as Schmidt coefficients.

Schmidt decomposition is a consequence of singular value decomposition when representing $|\psi\rangle$ as a tensor of order 2,

$$\Psi_{AB} = U \Lambda V^T, \quad (5)$$

where U and V are unitary matrices and Λ is a diagonal matrix.

Singular value decomposition works for rectangular matrices. As a consequence of this, Schmidt decomposition works for any bipartite quantum systems of different dimension, for example qubit-qutrit system.

As a convention, we usually require an ordering condition, i.e. $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$. This makes sure that the unitary matrices U and V to be uniquely determined.

For two qubits, the reduced density matrices ρ_A and ρ_B are

$$\rho_A = \begin{pmatrix} |\psi_{00}|^2 + |\psi_{01}|^2 & \psi_{00}\bar{\psi}_{10} + \psi_{01}\bar{\psi}_{11} \\ \bar{\psi}_{00}\psi_{10} + \bar{\psi}_{01}\psi_{11} & |\psi_{10}|^2 + |\psi_{11}|^2 \end{pmatrix}, \quad (6)$$

$$\rho_B = \begin{pmatrix} |\psi_{00}|^2 + |\psi_{10}|^2 & \psi_{00}\bar{\psi}_{01} + \psi_{10}\bar{\psi}_{11} \\ \bar{\psi}_{00}\psi_{01} + \bar{\psi}_{10}\psi_{11} & |\psi_{01}|^2 + |\psi_{11}|^2 \end{pmatrix}. \quad (7)$$

Question: Show that

$$\rho_A = \Psi_{AB} \Psi_{AB}^\dagger, \quad (8)$$

$$\rho_B = \Psi_{AB}^T \bar{\Psi}_{AB}. \quad (9)$$

As a result, when singular value decomposition is applied,

$$\rho_A = \Psi_{AB} \Psi_{AB}^\dagger = U \Lambda V^T \bar{V} \Lambda U^\dagger = U \Lambda^2 U^\dagger, \quad (10)$$

$$\rho_B = \Psi_{AB}^T \bar{\Psi}_{AB} = V \Lambda U^T \bar{U} \Lambda V^\dagger = V \Lambda^2 V^\dagger. \quad (11)$$

This shows that U and V are the eigendecomposition of the reduced density matrices ρ_A and ρ_B respectively, and the reduced density matrices ρ_A and ρ_B have the same eigenvalues (singular values or Schmidt coefficients).

For two qubits, there are two Schmidt coefficients λ_1 and λ_2 , and we know that $\lambda_1^2 + \lambda_2^2 = 1$. Therefore, there are only three possibilities,

- ① Maximally entangled states: $\lambda_1^2 = \lambda_2^2 = \frac{1}{2}$
- ② Entangled states: $\lambda_1^2 > \lambda_2^2 \neq 0$
- ③ Separable states: $\lambda_1^2 = 1, \lambda_2^2 = 0$

Question: Classify the following two-qubit states according to the framework.

- ① $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$
- ② $\frac{1}{\sqrt{3}}(|00\rangle + |01\rangle + |10\rangle)$
- ③ $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$

One can also describe the separability of two-qubit states by finding the separability condition. Let $|\psi_A\rangle \in \mathcal{H}_A$ and $|\phi_B\rangle \in \mathcal{H}_B$, and

$$|\psi_A\rangle = \psi_0|0\rangle + \psi_1|1\rangle, \quad (12)$$

$$|\phi_B\rangle = \phi_0|0\rangle + \phi_1|1\rangle. \quad (13)$$

The combined two-qubit state becomes

$$\begin{aligned} |\psi_{AB}\rangle &= |\psi_A\rangle \otimes |\phi_B\rangle \\ &= \psi_0\phi_0|00\rangle + \psi_0\phi_1|01\rangle + \psi_1\phi_0|10\rangle + \psi_1\phi_1|11\rangle. \end{aligned} \quad (14)$$

One can notice that the product of probability amplitudes of $|00\rangle$ and $|11\rangle$ is the same as the product of probability amplitudes of $|01\rangle$ and $|10\rangle$, when the two-qubit state is separable.

Therefore, the separability condition of two qubits is given by

$$\psi_{00}\psi_{11} - \psi_{01}\psi_{10}. \quad (15)$$

If $\psi_{00}\psi_{11} - \psi_{01}\psi_{10} = 0$, the two-qubit state is separable. Otherwise, it is entangled.

Last but not least, notice that the separability condition of two qubits is the same as the determinant of Ψ_{AB} ,

$$\Psi_{AB} = (\psi_{ij}) = \begin{pmatrix} \psi_{00} & \psi_{01} \\ \psi_{10} & \psi_{11} \end{pmatrix}. \quad (16)$$

The whole process is called entanglement classification of two qubits, which is well-studied for bipartite systems due to Schmidt decomposition.

From group theoretic perspective, the action of U and V is called local unitary group action on two qubits. Notice that U only acts on qubit A , V only acts on qubit B . Hence, singular value decomposition (or equivalently, Schmidt decomposition) on a two-qubit state can be written as

$$\begin{aligned}
 U \otimes V |\psi_{AB}\rangle &= \sum_{ij} U \otimes V \psi_{ij} |ij\rangle \\
 &= \sum_{ijklmn} \psi_{ij} |k\rangle \langle k|U|m\rangle \langle m| \otimes |l\rangle \langle l|V|n\rangle \langle n|(|i\rangle \otimes |j\rangle) \\
 &= \sum_{ijklmn} \psi_{ij} |k\rangle \langle k|U|m\rangle \langle m|i\rangle \otimes |l\rangle \langle l|V|n\rangle \langle n|j\rangle \\
 &= \sum_{ijkl} \psi_{ij} |k\rangle \langle k|U|i\rangle \otimes |l\rangle \langle l|V|j\rangle \\
 &= \sum_{ijkl} \psi_{ij} u_{ki} v_{jl} |k\rangle \otimes |l\rangle = \sum_{ijkl} u_{ki} \psi_{ij} v_{jl}^T |k\rangle \otimes |l\rangle \quad (17)
 \end{aligned}$$

Question: How many real parameters do you need to write down a 2×2 unitary matrix of determinant 1?

You need 3 real parameters to define a 2×2 unitary matrix,

$$U = \begin{pmatrix} u_{11} & u_{12} \\ -\bar{u}_{12} & \bar{u}_{11} \end{pmatrix}. \quad (18)$$

In order to specify a generic two-qubit state, you need 6 real parameters (ignoring the global phase).

Hence, it is possible for the local unitary group of two qubits to decompose a generic two-qubit state into the Schmidt coefficients, which are real and non-negative.

Question: Can we have Schmidt decomposition for three qubits?

State purification

Suppose we are given a mixed state ρ_A . It is possible to introduce another (reference) system, R , in such a way that the combined quantum state $|AR\rangle$ is a pure state, and $\rho_A = \text{Tr}_R(|AR\rangle\langle AR|)$.

Due to Schmidt decomposition, ρ_A can be written as

$$\rho_A = \sum_i p_i |i_A\rangle\langle i_A|. \quad (19)$$

The combined system needs to be in this form

$$|AR\rangle = \sum_i \sqrt{p_i} |i_A\rangle |i_R\rangle. \quad (20)$$

The partial trace Tr_R of the combined system is given by

$$\begin{aligned} \text{Tr}_R(|AR\rangle\langle AR|) &= \sum_{ij} \sqrt{p_i p_j} |i_A\rangle\langle j_A| \text{Tr}(|i_R\rangle\langle j_R|) = \sum_{ij} \sqrt{p_i p_j} |i_A\rangle\langle j_A| \delta_{ij} \\ &= \sum_i p_i |i_A\rangle\langle i_A| = \rho_A \end{aligned} \quad (21)$$